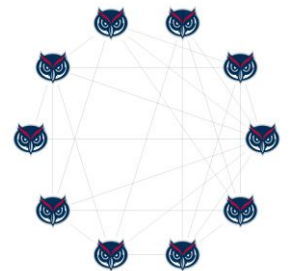




# Introduction to Network Analysis

Dr. Tom R. Leppard



## 1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

*R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.*

DOWNLOAD AND INSTALL R

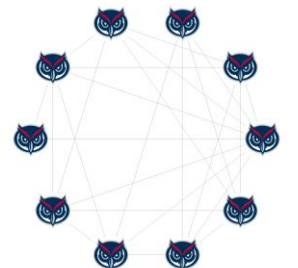
## 2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

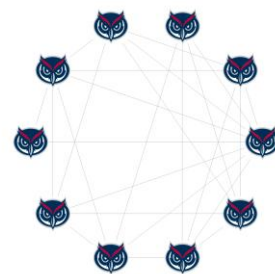
Size: 287.97 MB | [SHA-256: 8CE88C63](#) | Version: 2025.09.0+387 | Released: 2025-09-12

# R and R Studio – Download NOW

<https://posit.co/download/rstudio-desktop/>

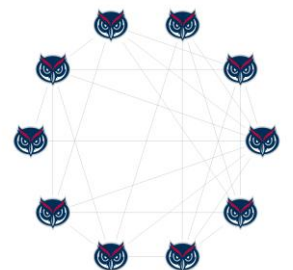


# Let's Meet

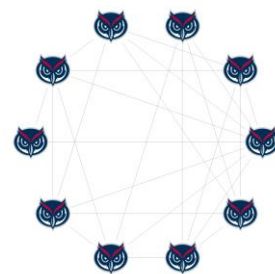


# What are Networks? What is Network Analysis?

- A system with individual interacting parts
- A visual representation of a system or group of people
  - Social, transportation, neural, epidemiological, cellular, commercial (travel/trade).
- A process of analysing individuals, groups, and/or systems using graph theory
  - Analyse observed, latent, and structural social processes.
- Building statistical models to explain why interactions occur, information spreads, or the prominence of people.

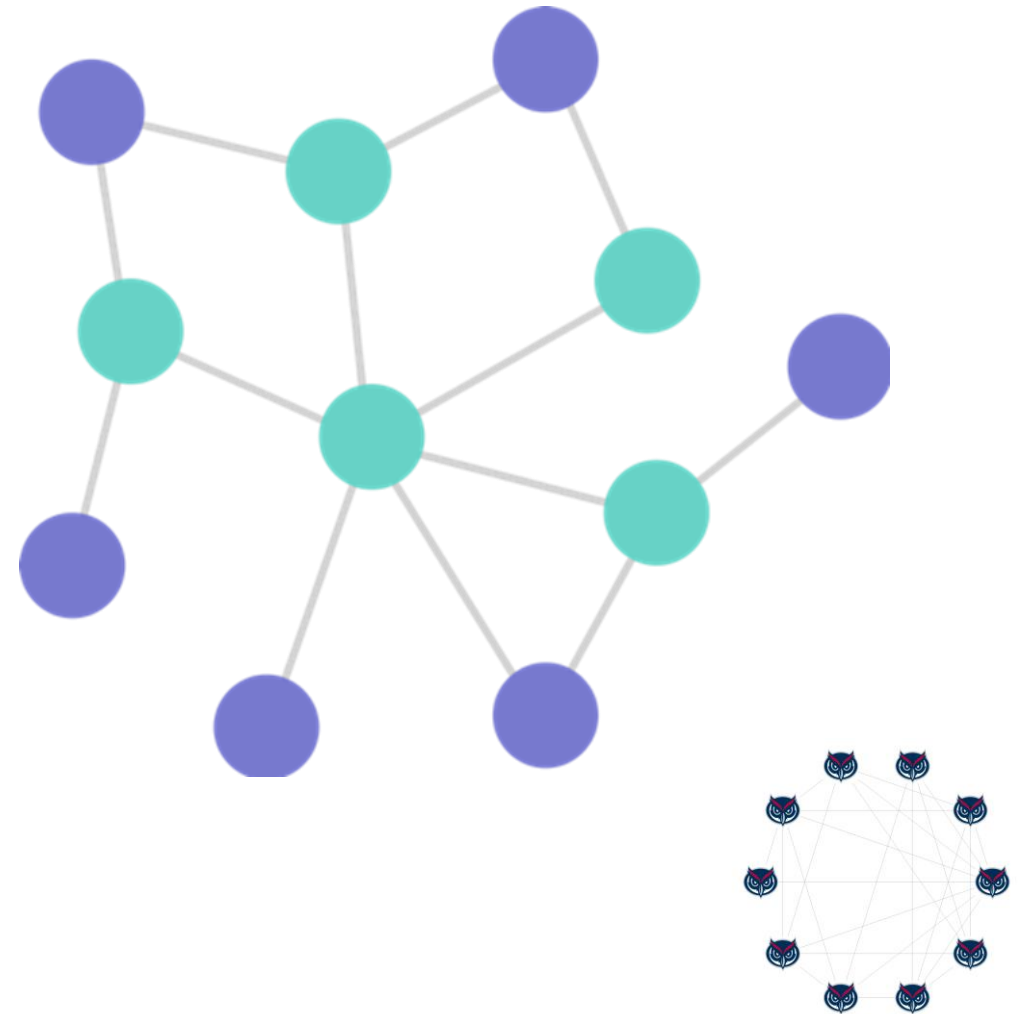


# Network Components



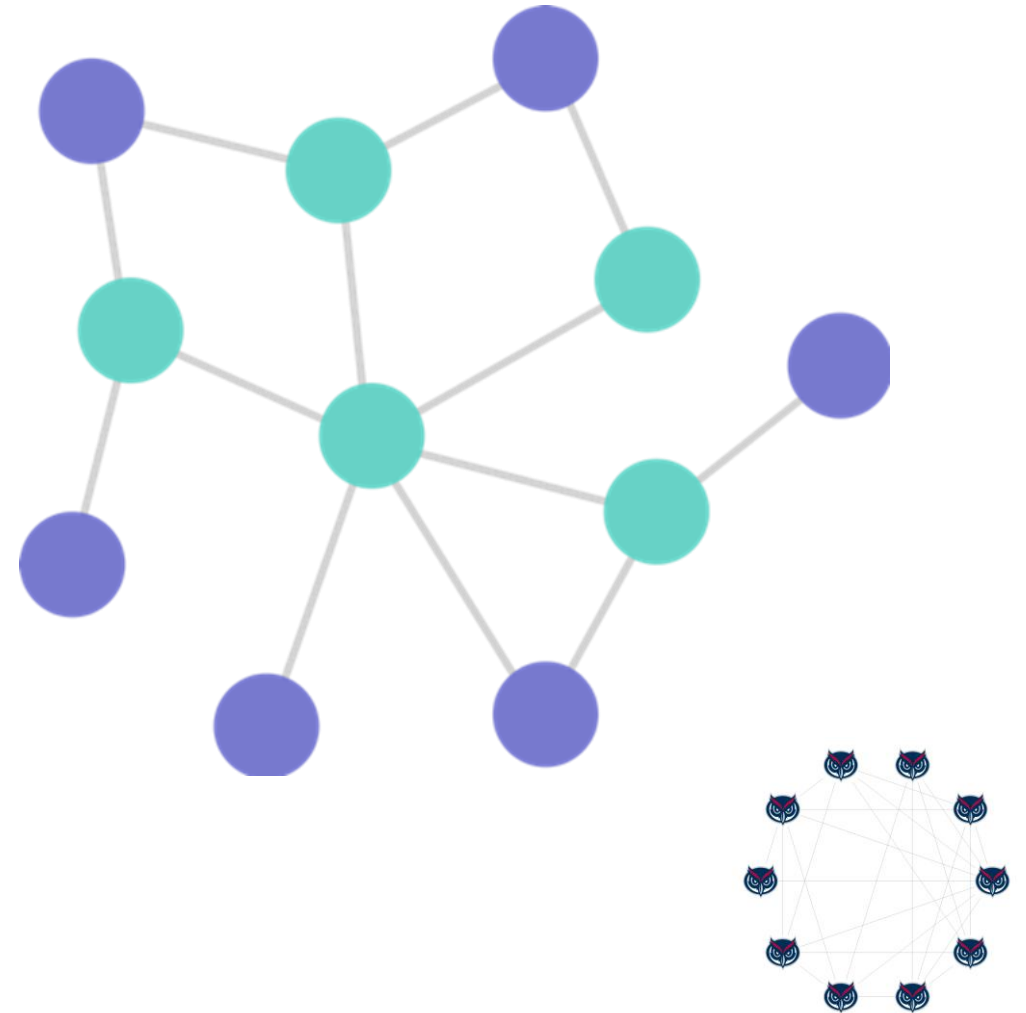
# Components - Nodes

- The nodes are the actors of a network.
  - People, groups, organizations, states, words, sentences, books, etc.
- The nodes in the figure are represented by the purple and green dots.

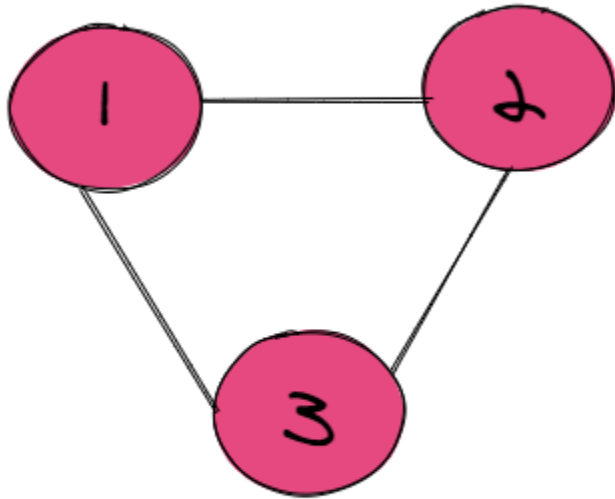


# Components - Edges

- Edges are the connections in the network.
  - Kinship, affiliation, interaction, affective relationships, etc.
- The edges are represented in the figure by the gray lines linking each node to another.



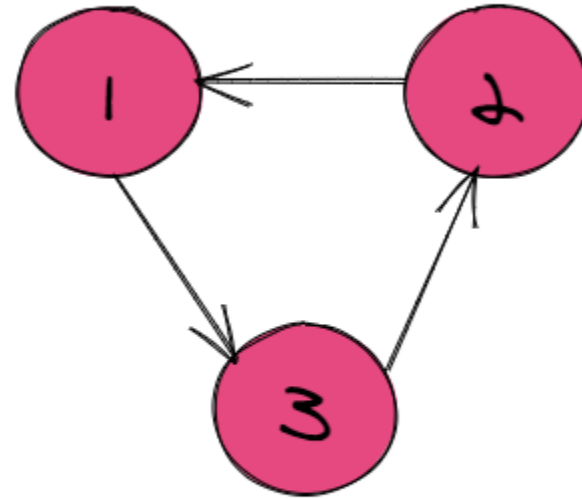
Undirected graph



Vertices =  $\{1, 2, 3\}$

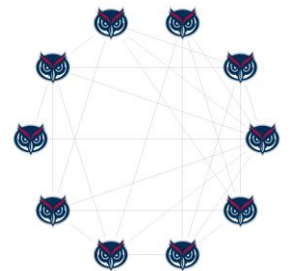
Edges =  $\{ \{1, 2\}, \{2, 3\}, \{3, 1\} \}$

Directed graph

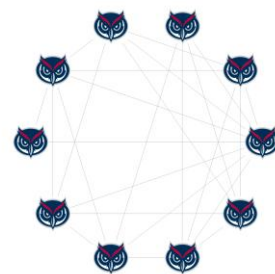


Vertices =  $\{1, 2, 3\}$

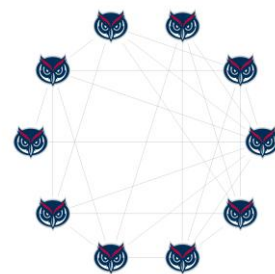
Edges =  $\{ \{2, 1\}, \{1, 3\}, \{3, 2\} \}$



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| Similarities                                                |                                                                 |                                                                   | Social Relations                                   |                                                                                   |                                                     |                                                                            | Interactions                                                            | Flows                                                             |
|-------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Location</b><br>e.g.,<br>Same spatial and temporal space | <b>Membership</b><br>e.g.,<br>Same clubs<br>Same events<br>etc. | <b>Attribute</b><br>e.g.,<br>Same gender<br>Same attitude<br>etc. | <b>Kinship</b><br>e.g.,<br>Mother of<br>Sibling of | <b>Other role</b><br>e.g.,<br>Friend of<br>Boss of<br>Student of<br>Competitor of | <b>Affective</b><br>e.g.,<br>Likes<br>Hates<br>etc. | <b>Cognitive</b><br>e.g.,<br>Knows<br>Knows about<br>Sees as happy<br>etc. | e.g.,<br>Sex with<br>Talked to<br>Advice to<br>Helped<br>Harmed<br>etc. | e.g.,<br>Information<br>Beliefs<br>Personnel<br>Resources<br>etc. |

**Fig. 3.** A typology of ties studied in social network analysis.